CS5228 DATA MINING PROJECT

# Predicting HDB monthly rent in Singapore

## CS5228 project progress report

October 2026

Internal out-of-time validation; no Kaggle test score reported

# Task, data and evaluation question

**Target** Monthly approved rent (SGD) for an individual HDB rental record.

**Labeled train** 150,000 rows, Jan 2021–Mar 2025.

**Unlabeled test** 50,000 rows, Mar 2025–Jul 2026; 49,620 postdate the final train month.

**Project questions**

- What do property, time and location explain?
- Do blocks and amenities improve prediction?
- Do COE and stocks add a reliable leading signal?

**Most test rows are future-dated: validation must move forward in time.**

Source: `reports/train_test_eda/summary.json`; CS5228 project brief (provided)

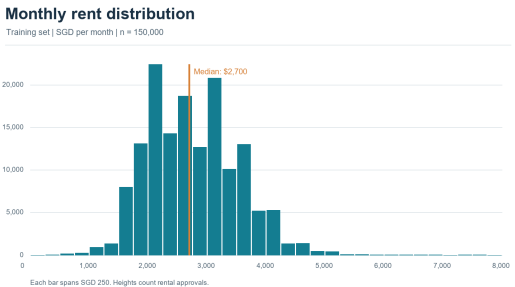
# Data audit: clean core table, but important caveats

| Check | Observed result and treatment |
|---|---|
| Missing/duplicates | No empty core cells; 425 exact train duplicates. Retain ambiguous copies because there is no rental ID to prove error. |
| Constant fields | `FURNISHED` is always yes; `FEE` is always zero. Exclude from learning. |
| Inconsistency | One lease-commencement year exceeds approval year; flag the resulting age rather than assuming it is valid. |
| Categories | Normalize labels such as `4 room` / `4-room`; fit categorical encoding on training months only. |
| Auxiliary sentinels | Treat block `NIL` and `-1` as missing, not physical values. |

The stock file also has missing quote fields; its data quality is separate from the complete rental table.

Source: `reports/train_test_eda/summary.json; reports/auxiliary_data_eda/summary.json`

# Property EDA: rent spread, size and flat type



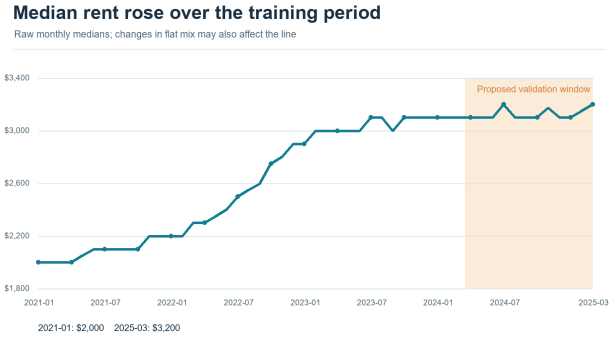

| Flat type | $n$ | Median SGD |
|---|---|---|
| 2-room | 3,098 | 2,100 |
| 3-room | 49,024 | 2,350 |
| 4-room | 54,079 | 2,800 |
| 5-room | 35,523 | 3,000 |
| Executive | 8,234 | 3,000 |

Area: 31–215 $m^2$; raw area–rent $r = 0.329$.
One-room: just 42 rows.

Rent: min 300; Q1 2,100; median 2,700; Q3 3,200; P95 4,000; P99 4,600; max 7,600 SGD. Raw associations can reflect other features.

Source: `reports/train_test_eda/summary.json`; `reports/train_test_eda/01_rent_distribution.png`

# Temporal EDA: a strong nonstationary level




Raw monthly median rent rose from SGD 2,000 (Jan 2021) to SGD 3,200 (Mar 2025). This is a 60% level increase, not necessarily a same-flat appreciation estimate.

**Month, recency and a forward holdout belong in the baseline.**

Source: `reports/train_test_eda/summary.json`; `reports/train_test_eda/02_monthly_median.png`

# Spatial EDA: town heterogeneity and block coverage

| Selected town | *n* | Median SGD |
|---|---|---|
| Bukit Timah | 438 | 3,000 |
| Central | 2,278 | 3,000 |
| Bishan | 3,306 | 2,900 |
| Punggol | 6,668 | 2,850 |
| Bedok | 8,907 | 2,500 |
| Yishun | 7,913 | 2,500 |

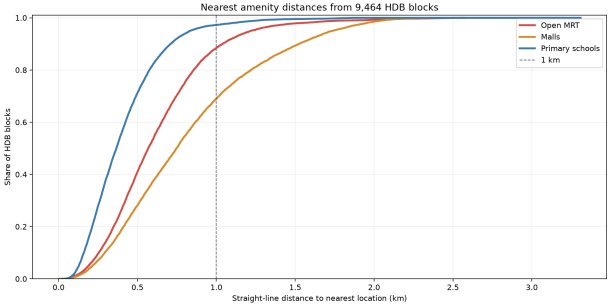
Across 26 towns, median rent ranges from SGD 2,500 to 3,000. Town sizes and flat mixes differ.

Normalized town–block–street keys match **100%** of train and test rows to 9,464 block records.

Coordinates, planning area, subzone, completion year and maximum floor add within-town detail.

Source: `reports/train_test_eda/summary.json; reports/auxiliary_data_eda/summary.json`

# Amenity EDA: accessible, but snapshot-based




| Nearest | Median km | $\leq$1 km |
|---|---|---|
| MRT | 0.568 | 88.46% |
| Mall | 0.747 | 68.97% |
| Primary school | 0.366 | 97.23% |

192 unique open MRT locations, 113 malls and 180 unique primary-school locations feed straight-line distance and count features.

Facility files are current snapshots: historical opening/closure dates are not available, so temporal interpretation is limited.

Source: `reports/auxiliary_data_eda/summary.json`; `reports/auxiliary_data_eda/02_nearest_distances.png`

# Economic associations have not yielded stable row-level gains

| Pair of monthly series | Transformation | Pearson $r$ |
|---|---|---|
| Raw mean rent vs COE C premium | Levels; COE lag 1 | $+0.960$ |
| Mix-adjusted rent vs COE C | Growth; COE lag 1 | $+0.306$ |
| Mix-adjusted rent vs SG stock | Growth/return; lag 1 | $-0.181$ |

Growth correlations use about 37 prevalidation months. The mix-adjusted rent index controls flat type/model, town, area and lease start; its adjustment is fitted before Apr 2024.

Against the same E baseline, adding engineered COE and SG-stock features together worsened row-level RMSE in all four forward windows by SGD 0.31–3.84. Individual economic sets sometimes helped a window slightly, but not consistently.

**Monthly association is not evidence of a stable forward prediction gain.**

Source: `reports/temporal_correlations/pair_correlations.csv`; `reports/economic_feature_ablation/forward_scores.csv`

# Grouped time series: descriptive signal, mixed forecasts

**Planning-area correlations**

Lagged COE C bids-to-quota ratio vs area rent growth, across 24 eligible areas:

| Rent statistic | Positive | Median $r$ |
|---|---|---|
| Mean | 20/24 | +0.148 |
| Median | 17/24 | +0.089 |

Correlation is modest; neither sign nor magnitude establishes a causal effect.

**Pooled panel Ridge, median-rent RMSE**

| Forward start | +COE | +stocks |
|---|---|---|
| Jan 2023 | +22.63 | +16.78 |
| Jul 2023 | +9.78 | +1.24 |
| Apr 2024 | −17.50 | −2.76 |

Deltas are SGD vs history-only; positive is worse. This one-step setup uses the *observed* prior-month rent, unlike batch test forecasting.

Source: `reports/group_temporal_panel/report.md`; `reports/group_temporal_panel/group_correlations.csv`

# Baseline protocol and leakage controls

- **Forward split:** fit Jan 2021–Mar 2024 (114,842 rows); hold out Apr 2024–Mar 2025 (35,158 rows).
- **Target and metrics:** record-level rent in SGD; compare RMSE, MAE, mean prediction bias, and month-level errors.
- **Seven core features:** flat type, model, area, age, approval month index, month-of-year, town.
- **Preprocessing:** normalize labels; learn imputation, scaling and categorical encoding from fit months only.
- **Ablation:** add block, amenity, COE and stock groups on identical rows, split and model settings.
- **Forecasting caveat:** one-month-updated signals differ from a fixed-origin forecast for all future test months.

Source: `reports/model_baseline/results.json; reports/model_families/results.json; reports/auxiliary_ablation/results.json`

# Baselines: feature groups change the answer

| Model / cumulative feature group | RMSE | MAE | Bias |
|---|---|---|---|
| Fit-period mean, no features | 846.75 | 690.60 | −531.44 |
| CatBoost A: type, model, area | 761.42 | 647.22 | −534.06 |
| CatBoost B: A + time/age | 529.96 | 397.86 | −49.91 |
| CatBoost C: B + town | **487.22** | **364.09** | −48.29 |

All scores are Apr 2024–Mar 2025 row-level SGD errors. CatBoost: 400 iterations, depth 6, learning rate 0.05, $L_2 = 3$, seed 42. No tuning on the test file.

**Time and age are added together in B; the A-to-B gain is not time's isolated contribution.**

Source: `reports/model_baseline/model_metrics.csv; reports/model_baseline/results.json`

# Model-family check on the same seven core features

| Model | RMSE | MAE | Bias |
|---|---|---|---|
| Ridge regression ($\alpha = 10$) | 624.88 | 468.18 | +361.99 |
| Random forest (160 trees, depth 18) | 493.77 | 370.19 | −56.65 |
| CatBoost (400 trees, depth 6) | 487.22 | 364.09 | −48.29 |
| **LightGBM (400 trees, 31 leaves)** | **485.09** | 361.41 | −42.68 |

Same 35,158-row holdout and seven features, with model-appropriate preprocessing. Ridge has a strong positive mean bias; trees handle nonlinear property/location effects better here.

**LightGBM leads CatBoost by only SGD 2.12 RMSE on this single split.**

Source: `reports/model_families/family_metrics.csv`; `reports/model_families/results.json`

# Auxiliary ablation: blocks and amenities help most

| HistGradientBoosting feature set | Features | RMSE | Bias |
|---|---|---|---|
| A: property | 3 | 758.51 | −531.85 |
| B: + time/age | 6 | 527.21 | −48.35 |
| C: + town | 7 | 485.72 | −46.78 |
| D: + block attributes | 15 | 480.22 | −46.90 |
| **E: + amenities** | 29 | **478.82** | −46.16 |
| F: + COE | 41 | 480.66 | −62.72 |
| G: + stocks | 47 | 481.31 | −65.93 |
| H: + interactions | 50 | 481.58 | −71.83 |

Same model (200 boosting iterations, 31 leaves, learning rate 0.05) and holdout throughout. E improves over C by SGD 6.89 RMSE (1.42%). F/G test the original macro feature sets; both worsen this split. A separate engineered-signal comparison also finds no consistent forward gain. Do not compare this ablation to LightGBM as a pure feature effect.

Source: `reports/auxiliary_ablation/ablation_metrics.csv`; `reports/economic_feature_ablation/forward_scores.csv`

# Diagnostics: useful signals, persistent drift

**E: block + amenities**

RMSE 478.82; MAE 356.36; mean bias −46.16 SGD. In Mar 2025, RMSE 457.45 but bias −80.10: average underprediction persists.

CatBoost C has RMSE 594.62 on 1,778 executive-flat holdout rows; only 14 one-room rows are too sparse for a stable slice.

**E permutation importance**

Largest MSE increases in a 2,000-row validation sample: flat type 151k, subzone 100k, area 31.6k; nearest mall 2.5k and MRT 2.3k.

Correlated features can share importance; this is predictive reliance, not an amenity price effect.

**More forward splits and month-/group-level error audits matter more than a tiny single-split gain.**

Source: `reports/auxiliary_ablation/feature_importance.csv`; `reports/auxiliary_ablation/monthly_errors.csv`; `reports/model_baseline/flat_type_errors.csv`

# Historical rent summaries improve every fixed-origin window

**Row-level RMSE by forecast origin**

| Start / horizon | Spatial | Historical | Change |
|---|---|---|---|
| Jan 2023 / 6 mo | 642.74 | 639.02 | −3.71 |
| Jul 2023 / 9 mo | 567.69 | 563.35 | −4.34 |
| Apr 2024 / 12 mo | 478.82 | 469.61 | −9.21 |

Pooled RMSE falls from **549.10** to **543.10**, a SGD 6.01 improvement. Signature, block–flat-type and area–flat-type summaries use only earlier labels and remain frozen after each origin.

**Spatiotemporal residual correction**

Planning-area × flat-type residual forecasts use group, seasonal and nearby-area lags. They reduce pooled RMSE by only SGD 0.27.

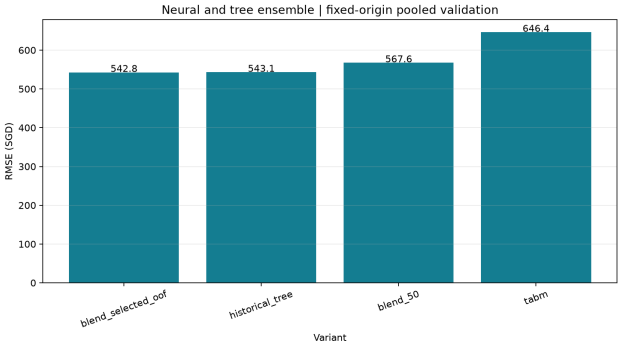
Combining the correction with historical features gives 543.13, slightly worse than historical features alone.

**Risk:** target histories need repeat observations. New properties require fallback estimates, and careless construction can leak future rent.

**Leakage-safe historical rent features provide the clearest new gain.**

Source: `reports/spatiotemporal_hybrid/fold_metrics.csv`; `reports/spatiotemporal_hybrid/pooled_metrics.csv`

# The neural model does not beat the historical tree




**TabM-style embedding ensemble**

Block and property embeddings help fit the training rows, but pooled fixed-origin RMSE is **646.37**. Mean bias is −133.30 SGD.

A 50/50 tree–neural blend scores 567.63. An out-of-fold-selected blend assigns 94% to the tree and 6% to the neural model, reaching **542.78**.

The SGD 0.31 gain is selected and measured on the same folds. It worsens the first fold by SGD 1.33, so it is not an unbiased improvement estimate.

**Keep the tree as the primary model. Treat the 6% neural blend as an experiment.**

Source: `reports/neural_tabm/pooled_metrics.csv; reports/neural_tabm/fold_metrics.csv`

# Trade-offs in row-level models and spatial features

| Method | Why it helps | Main limitation |
|---|---|---|
| Ridge | Fast, transparent baseline with standardized coefficients. | Misses nonlinear property effects and shows +SGD 362 bias on the core task. |
| Random forest | Handles nonlinearities with little preprocessing. | RMSE 493.77 trails boosting. Tree thresholds do not extend future rent trends. |
| Boosted trees | Best standalone row models with strong mixed-type handling. | Results vary by time split, and extrapolation beyond fitted time levels remains weak. |
| Block and amenities | Add local detail. E improves over town-only C by SGD 6.89. | Facility data are current snapshots and describe predictive proxies, not historical amenity effects. |

Source: `reports/model_families/family_metrics.csv; reports/auxiliary_ablation/ablation_metrics.csv`

# Trade-offs in temporal and neural extensions

| Method | Why it helps | Main limitation |
|---|---|---|
| Economic lags | Add interpretable market context. | About 37 independent prevalidation months exist. Tested variants show no stable gain. |
| Historical targets | Improve all three fixed-origin windows. | Need repeat entities and leakage controls. Unseen properties require fallback estimates. |
| Spatiotemporal residuals | Share nearby-area trends under a fixed origin. | Improve pooled RMSE by only SGD 0.27. Recursion can accumulate error. |
| TabM | Learns property and location embeddings. | Standalone RMSE is 646.37. Training is costly, and temporal bias is unstable. |
| 94/6 blend | Adds a small amount of neural error diversity. | The SGD 0.31 gain reuses the scoring folds and worsens the first fold. |

Source: `reports/economic_feature_ablation/report.md`; `reports/spatiotemporal_hybrid/report.md`; `reports/neural_tabm/report.md`

# Current conclusion and next experiments

**Current evidence**

Historical boosted trees are the primary candidate. Pooled fixed-origin RMSE is 543.10, and the latest 12-month fold reaches 469.61. Static block and amenity features remain useful. The standalone neural model and economic lags do not improve forward accuracy consistently.

**Next experiments**

1. Keep the historical tree primary until nested blend selection and multiple neural seeds confirm that the 94/6 blend is stable.
2. Improve extrapolation with detrended targets or direct horizon prediction. Keep target histories frozen at the forecast origin.
3. Retain only availability-safe external variables that improve several forward windows, then audit sparse and unseen groups before final refit.

Source: `reports/spatiotemporal_hybrid/report.md`; `reports/neural_tabm/report.md`